Introduction to Computational Chemistry & Biochemistry
Project 1: MM/MQ Basics

Write a (~1500-word) report, containing a brief introduction into the computational methods you used for the calculation (15%) with associated literature references (10%). Briefly summarise your results and comment on the accuracy of the numbers you have obtained (10%). Molecular drawings and plots of computational results are a critical aspect of the report and should be presented as publication quality figures with associated captions (50%). Include an appendix with detailed parameters needed to reproduce computational reported results, and details and acknowledgement of all uses of AI (15%).

Choose **one** of the following project topics.

1. **CHEMISTRY:** Difluorobutane ($C_4H_8F_2$) is a classic model for understanding stereoelectronic forces and the "gauche effect," as well as actively being researched as a potential low global warming potential refrigerant gas. Do a series of MM/MQ computations on the 6 distinct structural (constitutional) isomers based on straight-chain difluorobutane with the goal of determining which maximizes the gauche conformation at room temperature and pressure (STP). Summarize computational results with molecular visualizations and data plots as figures in your report along with tables. Also, provide a fundamental explanation of 'how and why' various isomers change the anti and gauche conformational populations. To understand and explain 'how and why' different straight-chain isomers of difluorobutane alter the populations of anti and gauche conformers, look at the interplay of four primary chemical phenomena captured by electronic structure calculations: (i) The fluorine gauche effect (stereo-electronics), (ii) electrostatic repulsion versus attraction, (iii) steric hindrance and atomic/molecular size, (iv) molecular thermodynamics (Boltzmann weighting).

Recommendations:

- Molecular Mechanics (MM) & Dynamics (MD) can be done directly in a jupyter/colab notebook using RDKit (MMFF94 force field) or OpenMM (calcflow). A recent and modern trend is to use ML developed force fields, MACE-OFF is recommended for molecular organic systems.
- Electronic Structure Calculations (QM) can be loaded into jupyter/colab notebooks using pySCF, but I would recommend using Gaussview/Gaussian16 (on ASU Sol or Calcflow). However, students are welcome to explore and use any common open-source QM software, e.g., ORCA, Psi4, GAMESS.
- Jupyter/Colab (or Marimo) notebooks are recommended for all computational data processing, analysis, plotting, visualization and combining everything into report figures.

2. **BIOCHEMISTRY:** A comprehensive understanding of how side-chain identity modulates backbone free-energy landscapes is fundamental to accurately modeling and understanding structure and dynamics of proteins. A comprehensive dataset mapping out the 20-canonical dipeptide sequence space establishes distinct rules for how amino acids populate their ϕ (phi) and ψ (psi) torsion angles. These energy landscapes and corresponding mapping out of distinct structure populations and molecular dynamics is largely dependent on the local environment (solvent, pH, ionic strength, temperature, osmotic pressure, etc). Hence, this is a really large combined configuration-environment space for computation and nearly intractable for experiment. Alanine dipeptide (typically modeled as N-acetyl-L-alanine N'-methylamide, Ac-Ala-NHMe) serves as the primary benchmark model in computational chemistry and biochemistry to study protein backbone folding and torsional dynamics because it is one of the simplest and features a well-distributed ϕ-ψ torsion angles map featuring helical, sheet and coil states. Altering the side chain introduces dramatic steric and electrostatic effects. Do a series of MM/MQ computations on the Alanine(A)/Phenylalanine(F) dipeptides (AA, AF, FA, FF) with the goal of determining how this one side-chain variation modulates backbone free-energy landscapes (Ramachandran plot) in a water solvent at room temperature and pressure (STP). Quality, stability, and convergence of MD simulations should be shown through thermodynamic and structural parameter plots (RMSD, RoG, etc.). Importantly, provide a fundamental explanation of 'how and why' various A/F dipeptides change the conformational populations. To understand and explain 'how and why' different amino acids alter the populations of common structural conformers (helical, sheet and coil), look at the interplay of four primary chemical phenomena captured by electronic structure calculations: (i) local steric hindrance (Van der Waals clashing), (ii) intramolecular electrostatic & quantum interactions (Hydrogen bonding & n-p orbital overlap), (iii) hyperconjugation and backbone orbital delocalization, (iv) sidechain-backbone and sidechain-sidechain π-interactions.

Recommendations:

- Molecular Mechanics (MM) & Dynamics (MD) should use protein specific force fields (e.g., AMBER, CHARMM, GROMOS) in standard open-source software like GROMACS or NAMD. Also, VMD is standardly used for MD visualization and parameter extraction.
- Electronic Structure Calculations (QM) can be done using either Gaussview/Gaussian16 (on ASU Sol or Calcflow). However, students are welcome to explore and use any common open-source QM software, e.g., ORCA, Psi4, GAMESS.
- Jupyter/Colab (or Marimo) notebooks are recommended for all computational data processing, analysis, plotting, visualization and combining everything into report figures.